

Yansheng Luo

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EDUCATION

University of Florida, Gainesville, FL

Expected May 2027

Bachelor of Science, Pre-professional Biology

Bachelor of Science, Chemistry (Biochemistry Emphasis)

Minors: Bioinformatics, Statistics

GPA: 3.99/4.00

Relevant Coursework: Biochemistry & Molecular Biology; Gene & Immunotherapy; Genomics & Biotechnology; Bioinformatics; Python Programming for Biology; Statistical Modeling; Stem Cell Biology; Virology.

RESEARCH PROFILE

Undergraduate researcher integrating computational genomics, statistical modeling, and experimental biology to investigate tumor heterogeneity, therapeutic response, and resistance. Experience spans clonal lineage tracing, transcriptomic meta-analysis, quantitative proteomics, and AI-driven computational protein design, with additional experience in regenerative biology. Current interests center on translational computational oncology, precision medicine and biomarker discovery, cancer genomics and multi-omics, and computational-experimental integration.

RESEARCH EXPERIENCE

The University of Texas MD Anderson Cancer Center

Jun 2026 – Aug 2026

Summer Research Trainee, ITERT/CATALYST Program – Translational Molecular Pathology

Advisor: Wantong Yao, PhD | Mentor: Yue Zeng

- Investigated treatment-history-dependent clonal evolution and resistance to KRAS-targeted therapy in pancreatic ductal adenocarcinoma (PDAC), comparing pharmacologic KRAS G12D inhibition, genetic KRAS extinction, sequential treatment, and rechallenge states to evaluate how therapeutic pressure and treatment history shape clonal bottlenecks, recurrence, and routes to resistance.
- Designed a clone-prioritization framework integrating recurrence, abundance, effect size, treatment specificity, trajectory patterns, and robustness to identify biologically meaningful resistance-associated subclones from a raw barcode universe exceeding 400,000 sequences.
- Built reproducible R workflows for clonal lineage tracing, differential abundance analysis, dimensionality reduction, clonal-composition analysis, trajectory comparison, sensitivity analysis, and robustness testing.
- Integrated clonal lineage-tracing results with quantitative proteomic profiling of human PDAC parental, MRTX-sensitive, and acquired-resistant states (6,851 unique proteins) to examine molecular programs associated with treatment response and resistance.
- Performed extensive mouse genotyping and PCR-based genotype validation, contributed to PDAC cell line establishment, and conducted multi-organ dissections, MRI tumor imaging, and tumor assessment in mouse models, linking computational analyses with in vivo phenotypes.
- Presented the integrated clonal-evolution study as a poster and lightning talk at MD Anderson Cancer Center.

University of Florida – Song Lab

Jan 2026 – Present

Undergraduate Researcher – Portable Innate Immune Response Score (IMRS)

Advisor: Qianqian Song, PhD

- Proposed and developed IMRS, a transferable transcriptomic scoring framework for quantifying acute delivery-associated innate immune activation across independent viral-vector and nanoparticle datasets without dataset-specific model retraining.
- Designed a frozen-weight meta-analysis framework combining differential expression, heterogeneity filtering, inverse-variance weighting, and power-controlled gene selection to score independent datasets without model retraining.
- Established a statistical validation framework using Cochran's Q, I^2 , power estimation, permutation testing, leave-one-anchor-out validation, sensitivity analysis, and contribution auditing to assess robustness and generalizability.
- Evaluated cross-species transferability by mapping mouse-derived response weights to human ortholog datasets for translational evaluation.
- Built reusable bulk RNA-seq pipelines in R and Python and benchmarked IMRS against baseline immune-response signatures across independent datasets, emphasizing frozen-weight portability without dataset-specific retraining.
- Audited gene-level contributions and score stability under gene-removal and threshold perturbations to identify dependence on individual features and characterize potential failure modes.

University of Florida – Cancer and Genetic Research Complex

Sep 2023 – Present

Undergraduate Researcher – Regeneration & Disease Biology

Advisor: Malcolm Maden, PhD

- Investigate mechanisms of peripheral nerve regeneration and fibrosis resistance using the African spiny mouse (*Acomys cahirinus*) as a comparative mammalian model of injury and repair.
- Designed and executed sciatic nerve injury experiments in *Acomys* and *Mus musculus* to compare regenerative and fibrotic responses through histological and molecular analyses.
- Performed H&E, Masson's trichrome, and immunohistochemistry on >20 tissue samples to characterize extracellular matrix remodeling, cellular injury responses, and fibrosis-associated markers.
- Developed and optimized whole-mount immunostaining protocols by systematically adjusting permeabilization/solvent conditions and incubation parameters to improve antibody penetration and staining reproducibility.
- Established and maintained >50 primary mammalian culture preparations, including dorsal root ganglion (DRG) explant cultures, to compare neurite-growth architecture and regenerative phenotypes across *Acomys* and *Mus*.
- Quantified DRG explant morphology using ImageJ, QuPath, and statistical analysis in R, identifying a distinct radially organized neurite-growth architecture in adult *Acomys* relative to *Mus* and supporting hypothesis refinement around species-specific regenerative phenotypes.

University of Florida – International Genetically Engineered Machine Team (iGEM)

Mar 2025 – Present

Dry Lab Subleader, AI-Driven Computational Protein Design

- Led expansion of the team's computational pipeline for designing target-binding proteins, integrating BindCraft with RFdiffusion backbone generation and ProteinMPNN sequence design on HiPerGator.
- Developed a quantitative filtering and ranking framework integrating structural confidence, interface quality, steric clashes, solvent exposure, sequence properties, and cross-model agreement to prioritize candidates for experimental testing.
- Stress-tested ranking robustness across alternative weighting schemes and computational runs using correlation, sensitivity, and rank-stability analyses, then calibrated the ranking strategy against AMBER molecular dynamics-derived stability and interaction metrics.
- Diagnosed chain-assignment errors and missing interface metrics in the RFdiffusion-to-ProteinMPNN/BindCraft workflow across the campaign's RFdiffusion backbone inputs, resolving issues that affected downstream filtering and ranking.
- Coordinated computational work across dry-lab, wet-lab, and human-practices groups; assigned targeted analyses, reviewed results, integrated findings into experimentally actionable candidate-selection decisions and project communication, and mentored newer dry-lab members through workflow walkthroughs and feedback.

Dry Lab Member, AI-Driven Computational Protein Design

Mar 2025 – Oct 2025

- Configured and optimized BindCraft workflows on HiPerGator for the team's initial osteocalcin binder-design campaign.
- Developed an automated Python K-means clustering workflow to analyze >700 RMSD datasets and support objective structural candidate selection.
- Built a HADDOCK3 docking workflow that narrowed >500 osteocalcin binder candidates to 10 candidates for downstream wet-lab validation; the team subsequently selected three top-ranked sequences for cloning, expression/purification, and biolayer interferometry (BLI), with measurable osteocalcin-binding activity reported in the public project poster.

University of Florida – Song Lab

May 2025 – Jul 2025

Undergraduate Researcher – Bioinformatics SNV Project

- Analyzed single-cell cancer sequencing data to characterize mutational patterns and intratumoral heterogeneity.
- Built a reproducible HPC workflow spanning FASTQ processing, BWA alignment, variant calling, and downstream analysis; processed >20 single-cell datasets using SCcaller and CellSNP-lite.
- Quantified and visualized mutation burden and variant distributions in R to support biological interpretation across tumor cell populations.

Additional Research Experience: Advanced Molecular Biology Lab Project (Fall 2025) – cloned and analyzed the TEM-1 β -lactamase gene using DNA purification, restriction digest, ligation, and PCR verification. Special Topics in Biotechnology (Spring 2024) – applied PCR, sequencing, and bioinformatics to regenerative gene-expression analysis in *Mus musculus* after ear injury.

PUBLICATIONS & MANUSCRIPTS

- Luo, Y., & Maden, M.** “Adult *Acomys* DRG Explants Exhibit a Radially Organized Neurite Architecture Distinct from *Mus* Explants.” University of Florida Journal of Undergraduate Research. Accepted for publication, 2026.
- Luo, Y., & Song, Q.** “A Frozen Transcriptomic Framework for a Shared Acute Delivery-Associated Innate Response Axis in Public Bulk RNA-seq Data.” Manuscript prepared for submission.
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RESEARCH PRESENTATIONS & SELECTED OUTPUTS

- Luo, Y., Zeng, Y., Attanasio, S., et al.** (Aug 2026). Different Routes to Resistance: Multifaceted KRAS Inhibition Directs Distinct Clonal Evolution in Pancreatic Ductal Adenocarcinoma. Poster presentation and lightning talk, MD Anderson Cancer Center, Houston, TX.
- Luo, Y., & Maden, M.** (Nov 2025). *Acomys* vs. *Mus*: Comparative Regeneration Revealed Through 3D DRG Cultures and Sciatic Nerve Injury Models. Poster, UF Department of Biology Undergraduate Honors & Research Symposium, Gainesville, FL.
- Luo, Y., & Nicolas, B.** (Sep 2025). Leveraging AI in Synthetic Biology: AI-Derived Enzyme Engineering on Osteocalcin. Oral presentation, AI at UF Forum, Gainesville, FL.
- Luo, Y., & Maden, M.** (Apr 2025). Regenerative Potential of the African Spiny Mouse (*Acomys cahirinus*): Insights into Peripheral Nerve Tissue Repair and Fibrosis Resistance. Poster, APSA x CURBS Research Symposium, University of Florida, Gainesville, FL.
- Luo, Y., & Maden, M.** (Apr 2025). Regenerative Potential of the African Spiny Mouse (*Acomys cahirinus*): Insights into Peripheral Nerve Tissue Repair and Fibrosis Resistance. Poster, University of Florida Spring Research Symposium, Gainesville, FL.
- UF iGEM Team. (2025). [Generative Design of Osteocalcin Binder Protein for Point-of-Care Based Sensor](#). Publicly archived research poster; undergraduate team member.
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LABORATORY & COMPUTATIONAL SKILLS

- **Programming & Statistics:** R, Python, MATLAB; statistical inference, multivariate analysis, differential expression analysis, meta-analysis, dimensionality reduction, clustering, permutation testing, sensitivity analysis, and robustness evaluation.
 - **Genomics & Bioinformatics:** Bulk RNA-seq; single-cell variant analysis; BWA, SCcaller, CellSNP-lite; FASTQ/BAM workflows; cross-species ortholog mapping; HPC/Slurm workflows.
 - **Structural Bioinformatics & Protein Design:** BindCraft, RFdiffusion, ProteinMPNN, HADDOCK3, AMBER molecular dynamics; structural confidence and interface metric analysis; candidate filtering, ranking, and robustness evaluation.
 - **Imaging & Quantification:** ImageJ, QuPath; MRI tumor imaging and assessment; quantitative histology and immunohistochemistry image analysis; phenotype extraction and comparative image quantification.
 - **Molecular & Cellular Biology:** Mammalian cell culture and cell line establishment; mouse genotyping and multi-organ dissection; cryosectioning; H&E, Masson's trichrome, IHC, whole-mount immunostaining; PCR, primer design, cloning, restriction digestion, ligation, sequencing; protein purification, chromatography, SDS-PAGE, Western blotting, ELISA, kinetic assays.
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HONORS & AWARDS

- Outstanding Undergraduate International Student Award, University of Florida College of Liberal Arts and Sciences – one undergraduate recipient per college; \$1,000 award (Mar 2026).
- University Scholar Program, University of Florida – awarded in two consecutive cycles; \$3,500 total undergraduate research funding (Apr 2024 – Apr 2026).
- International Student Achievement Awards (ISAA), Certificate of Outstanding Merit, University of Florida (Nov 2024).
- The Honor Society of Phi Kappa Phi – Elected member (Fall 2025).
- iGEM Grand Jamboree — Silver Medal, University of Florida iGEM Team (2025).
- President's Honor Roll, University of Florida (Fall 2023, Fall 2024, Spring 2025, Fall 2025); Dean's List (Spring 2024).

TEACHING & MENTORING

University of Florida | *Teaching Assistant – Intro Bioinformatics (BSC4434C)* *Aug 2025 – Dec 2025*

- Provided individualized instruction in bioinformatics concepts and computational tools and supported students with coursework and computational troubleshooting.

University of Florida | *Teaching Assistant – Microbiology Lab (MCB3020L)* *Jan 2025 – May 2025*

- Instructed students in Köhler illumination, Gram staining, organism identification, and ELISA; graded laboratory assignments and examinations.

University of Florida | *Teaching Assistant – Comparative World Agriculture (AEB4905)* *Jan 2024 – May 2024*

- Led weekly discussions, mentored students, graded coursework, and delivered a guest lecture on aquaponics and food security.

Community STEM Tutoring | *STEM Tutor & Mentor* *Oct 2020 – Present*

- Delivered 200+ hours of individualized STEM instruction to 30+ high school and undergraduate students in the United States, Australia, New Zealand, and China through virtual and in-person tutoring.
- Designed individualized lesson plans, problem sets, and review materials across biology, chemistry, biochemistry, anatomy, calculus, algebra, and statistics, adapting instruction to students' academic level, curriculum, and learning needs.
- Supported preparation for course examinations and standardized assessments across multiple curricula, tailoring problem-solving explanations to individual student needs.

PROFESSIONAL DEVELOPMENT & SPECIALIZED TRAINING

- Polyploidy Undergraduate Research Experience (PURE), virtual (Jun 29 – Jul 1, 2026) – genetics, evolution, ecology, polyploidy methods, and team science.
- NVIDIA Fundamentals of Deep Learning Workshop, University of Florida – hands-on neural networks, computer vision, transfer learning, and GPU-accelerated workflows.
- Clinical Perspectives for a Basic Scientist – MD Anderson/UTHealth Houston GSBS, ITERT Nanocourse (2026) – two-week graduate-level intensive training in translational oncology, including lung cancer, melanoma, neuro-oncology, mouse models, cancer pharmacology, therapeutic strategies, radiology, disease progression, metastasis, and survivorship.
- ITERT Professional Development Workshop Series, MD Anderson Cancer Center (2026) – training in critical appraisal, experimental planning, scientific reporting, communication, networking, and career development.

LEADERSHIP, SERVICE & ACTIVITIES

- 2026 ITERT Science Fair – taught K-12 students polymer chemistry through a sodium alginate/calcium chloride “science boba” demonstration.
- Herbert Wertheim College of Engineering Outreach Design Competition – designed a tabletop engineering outreach activity for middle-school students.
- National Honor Society, Palmer Trinity School – President (Oct 2022 – May 2023); directed school-wide initiatives and increased member engagement by 20%.
- Florida Immigrant Coalition – Student Captain (Feb 2020 – May 2023); coordinated community events and workshops and completed leadership training in project management and advocacy.
- NeuroGenetics Student Organization, University of Florida (Jan 2025 – May 2026) – Science Olympiad volunteer.
- Master Scuba Diver (NAUI Certified) – advanced scuba and emergency first-aid training.